Faculty of Applied Science and Technology

LAB #3 – Applications of Array Operations

Michaël McCorkell

Dr. Maryam Davoudpour

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Procedures and results: Add the number of all questions in this report, even if you decided to not answer them.

For each question, you should explain your observation + code snippet.

9

a)

```
>> u= [4  13 -7]

u =

     4    13    -7
```

b)

```
>> Lu =sqrt(sum(u.^2))

Lu =

   15.2971
```

c)

```
>> un=(u/Lu)

un =

    0.2615    0.8498   -0.4576
```

d)

```
>> un=sqrt(sum(un.^2))

un =

    1.0000
```

10-

```
>> u1 = [3.2 -6.8 9]; u2 = [-4 2 7];
>> angle = acosd(sum(u1 .* u2) / (sqrt(sum(u1.^2)) * sqrt(sum(u2.^2))))

angle =

   67.9273
```

11-

```
>> d= [2 4 3]

d =

     2     4     3
```

a)

```
>> d+d

ans =

     4     8     6
```

b)

```
>> d.^d

ans =

     4   256    27
```

c)

```
>> d.*d

ans =

     4    16     9
```

d)

```
>> d.^2

ans =

     4    16     9
```

12.

```
>> v=[3 -1 2]

v =

     3    -1     2

>> u=[6 4 -3]

u =

     6     4    -3
```

a)

```
>> v.*u

ans =

    18    -4    -6
```

b)

```
>> v.^u

ans =

  729.0000    1.0000    0.1250
```

c)

```
>> v*u'

ans =

     8
```

13.

```
>> v=[1 3 5 7]

v =

     1     3     5     7
```

a)

```
>> a = 3.*v

a =

     3     9    15    21
```

b)

```
>> b = v.^2

b =

     1     9    25    49
```

c)

```
>> c = 1.^v

c =

     1     1     1     1
```

d)

```
>> (v./v)*6

ans =

     6     6     6     6
```

14.

```
>> v = [5 4 3 2]

v =

     5     4     3     2
```

a)

```
>> a= 1./(v+v)

a =

    0.1000    0.1250    0.1667    0.2500
```

b)

```
>> b=v.^v

b =

        3125         256          27           4
```

c)

```
>> c=v./sqrt(v)

c =

    2.2361    2.0000    1.7321    1.4142
```

d)

```
>> (v.^2)./(v.^v)

ans =

    0.0080    0.0625    0.3333    1.0000
```

21.

```
>> g=9.81; v0=260; theta= [15 25 35 45 55 65 75];
>> s= ((v0^2)/g).*(sind(2*theta))

s =

   1.0e+03 *

    3.4455    5.2788    6.4754    6.8909    6.4754    5.2788    3.4455
```

23.

a)

```
>> n_a = 1:5; terms_a = (n_a.^2) ./ (2.^n_a); sum_a = sum(terms_a)

sum_a =

    4.4062
```

b)

```
>> n_b = 1:15; terms_b = (n_b.^2) ./ (2.^n_b); sum_b = sum(terms_b)

sum_b =

    5.9911
```

c)

```
>> n_c = 1:30; terms_c = (n_c.^2) ./ (2.^n_c); sum_c = sum(terms_c)

sum_c =

    6.0000
```

25.

```
>> format long

x = [pi/3 - 0.1, pi/3 - 0.01, pi/3 - 0.0001, pi/3 + 0.0001, pi/3 + 0.01, pi/3 + 0.1];
y = sin(x - pi/3) ./ (4 * cos(x).^2 - 1);

limit_val = -sqrt(3) / 6;

% Display results
disp('x values:');
disp(x.');

disp('y values:');
disp(y.');

disp('Exact limit value:');
disp(limit_val);
x values:
   0.947197551196598
   1.037197551196598
   1.047097551196598
   1.047297551196598
   1.057197551196598
   1.147197551196598

y values:
  -0.274238398404589
  -0.287032331771222
  -0.288658470333254
  -0.288691803667625
  -0.290366054030767
  -0.307964386863987

Exact limit value:
  -0.288675134594813
```

[redacted]

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Both sides of the sheet may be used;
Must be printed on 8.5" x 11" paper.

Course: **MENG2020** - ONA

Student's name: Michaël M

Student's signature: MichaëlM

n1=[10 14 8 6]
n2=[6
4
12
2]

n1=[100:100:400]
n1=100 200 300 400

n1 = [10:20:60]
n1= 10 30 50

n2= 'This is for fun?'
= This is for fun?

Variable Name	Variable Value
m	16 18 20 8
n	4 8 2 0 6 10 4 2 -2 2 -4 -6 8 12 6 4
p	200 400 600 800
q	200 400 600 800
r	60 84 48 36 40 56 32 24 120 168 96 72
s	60 56 96 12
t	400
u	4
v	16
w	3

Now complete the same three exercises using MATLAB. Are the results the same as expected?
If not, please record the discrepancies and explain how MATLAB execute the code that way.

After completing the three exercises in MATLAB, the results were consistent with those obtained from my written calculations. Each exercise produced the expected numerical outcomes, confirming the validity of the manual approach. While the overall results matched, I found the last training exercise required more careful consideration. This is because MATLAB strictly follows its own rules for order of operations, vectorization, and matrix handling, which sometimes differs from how we reason through problems by hand. Despite this adjustment, the outcomes remained the same, and the process reinforced my understanding of both the manual calculations and MATLAB's execution.

```
>> n1 = [10 14 8 6]
n2 = [6 4 12 2]'
m = n1' + n2
n = n1 - n2
r = n1 .* n2
s = n1 .* n2'

n1 =

    10    14     8     6


n2 =

     6
     4
    12
     2


m =

    16
    18
    20
     8


n =

     4     8     2     0
     6    10     4     2
    -2     2    -4    -6
     8    12     6     4


r =

    60    84    48    36
    40    56    32    24
   120   168    96    72
    20    28    16    12


s =

    60    56    96    12
```

```
>> n1 = [100:100:400]
p = n1' + n1'''
q = n1' + n1'
[t, u] = max (n1)

n1 =

   100   200   300   400


p =

   200
   400
   600
   800


q =

   200
   400
   600
   800


t =

   400


u =

     4
```

```
>> n1 = [10:20:60]
n2 = 'This is for fun?'
v = length (n2)
w = length (n1)

n1 =

    10    30    50


n2 =

    'This is for fun?'


v =

    16


w =

     3
```

References:

List all sources cited in the report with proper formatting in case of the use
MATLAB textbook : MATLAB: An Introduction with Applications 6th Edition
MATLAB Onramp : https://matlabacademy.mathworks.com/details/matlab-onramp/gettingstarted